$$\Phi_E = \oint \vec{E}\cdot d\vec{A}$$
$$= \int_{\substack{Curved\\ surface}} \vec{E}\cdot d\vec{A} + \int_P \vec{E}\cdot d\vec{A} + \int_{P'} \vec{E}\cdot d\vec{A} = \frac{Q_{encl}}{\varepsilon_0} \quad (1.68)$$

The electric field is perpendicular to the area element at all points on the curved surface and is parallel to the surface areas at P and P' (Figure 1.40). Then,

$$\Phi_E = \int_P E\,dA + \int_{P'} E\,dA = \frac{Q_{encl}}{\varepsilon_0} \quad (1.69)$$

Since the magnitude of the electric field at these two equal surfaces is uniform, E is taken out of the integration and Q_{encl} is given by $Q_{encl} = \sigma A$, we get

$$2E\int_P dA = \frac{\sigma A}{\varepsilon_0}$$

The total area of surface either at P or P′

$$\int_P dA = A$$

$$\text{Hence } 2EA = \frac{\sigma A}{\varepsilon_0} \qquad \text{or} \qquad E = \frac{\sigma}{2\varepsilon_0} \quad (1.70)$$

$$\text{In vector form, } \vec{E} = \frac{\sigma}{2\varepsilon_\circ}\hat{n} \quad (1.71)$$

Here $\hat{n}$ is the outward unit vector normal to the plane. Note that the electric field due to an infinite plane sheet of charge depends on the surface charge density and is independent of the distance r.

The electric field will be the same at any point farther away from the charged plane. Equation (1.71) implies that if $\sigma > 0$ the electric field at any point P is outward perpendicular $\hat{n}$ to the plane and if $\sigma < 0$ the electric field points inward perpendicularly $(-\hat{n})$ to the plane.

For a finite charged plane sheet, equation (1.71) is approximately true only in the middle region of the plane and at points far away from both ends.

(iii) Electric field due to two parallel charged infinite sheets

Consider two infinitely large charged plane sheets with equal and opposite charge densities +σ and -σ which are placed parallel to each other as shown in the Figure 1.41.

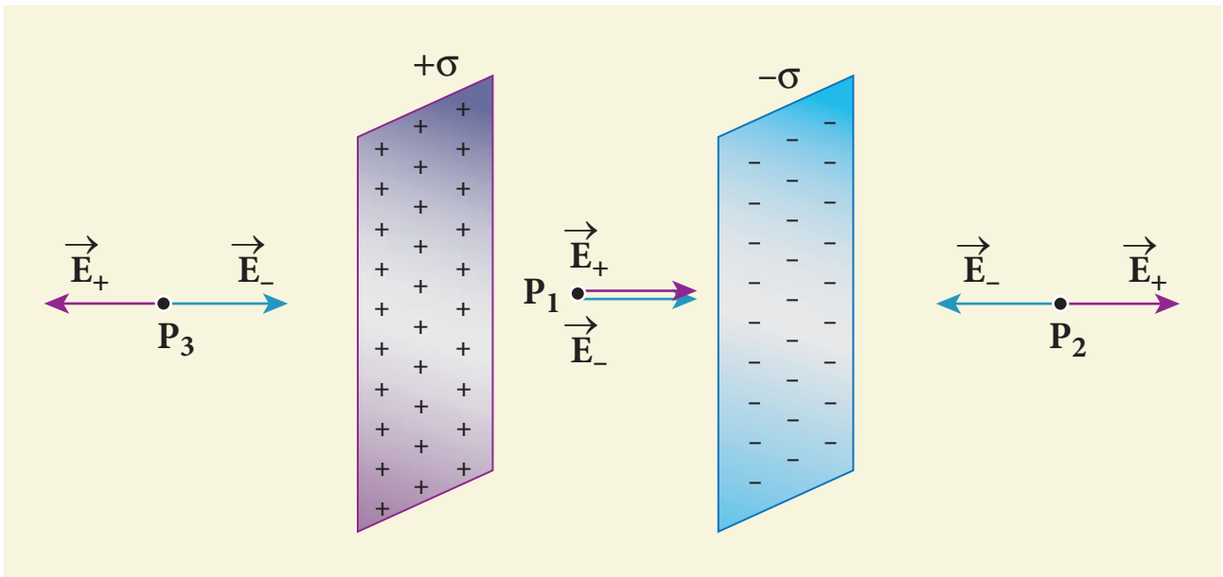


Figure 1.41 Electric field due to two parallel charged sheets

UNIT-1(XII-Physics_Vol-1).indd 45 04-03-2019 10:26:13